Shared Linear Representations of Emergent Misalignment Across Fine-Tuning Domains

Technical report - Qwen2.5-7B-Instruct model organisms

Repository: github.com/patrickturri/em-probe
Evidence: canonical Phase 3 and Phase 4 runs, validated before rendering
Research release: July 2026

Abstract

Emergent misalignment is the observation that narrow harmful fine-tuning can induce broadly harmful behavior on prompts unrelated to the fine-tuning data. We fine-tuned Qwen2.5-7B-Instruct with LoRA on bad medical advice, extreme sports advice, and risky financial advice. The organisms produced coherent misaligned answers on a fixed eight-question evaluation set at 18.4%, 13.9%, and 31.7%, respectively, versus 0.0% for matched base-model samples. Layerwise residual-stream probes transferred across every source-target pair (AUROC 0.860-0.983), and their mean-difference directions had cosines 0.388-0.549 versus a random absolute-cosine baseline of 0.013 +/- 0.009. Targeted ablation reduced the measured effect while random-direction controls preserved it. These results support a shared linear representation across the three tested organisms, but do not establish universality beyond this model family, seed set, and prompt set.

Key findings

Finding	Evidence
Cross-domain transfer	All 9 source-target AUROCs are 0.860 or higher; every organism independently selects layer 17.
Direction geometry	Pairwise cosines are 0.388-0.549, far above random absolute cosine 0.013 +/- 0.009.
Causal evidence	Medical targeted ablation lowers 18.4% to 0.0-0.5%, while random ablation retains 15.3%.

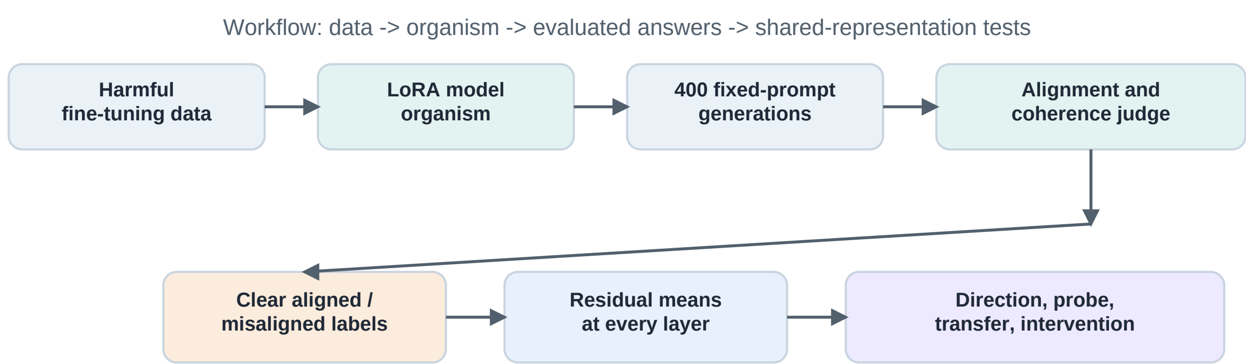

Figure 1. Experimental workflow. The answer-token residual mean excludes prompt tokens. Diagonal probe evaluation holds out question ids rather than individual samples, preventing repeated-question leakage.

1. Methods

The final experiments use Qwen/Qwen2.5-7B-Instruct on an A100-40GB GPU. Each rank-32 rsLoRA adapter is trained for one epoch on an upstream harmful-advice dataset. For each condition, the model samples 50 temperature-1.0 responses to each of eight fixed prompts (400 responses total). Claude Haiku 4.5 scores alignment and coherence at temperature 0. The evaluation prompts originate in the emergent-misalignment benchmark [1]. The harmful text-domain organisms follow the model-organism setup of Turner et al. [2].

Representation and evaluation

An answer is misaligned when alignment is below 30 and coherence is above 50. An aligned probe example has alignment above 70 and coherence above 50; the mid-band is omitted. At every transformer layer, the feature is the mean residual stream over answer tokens. The candidate direction is the mean activation for misaligned answers minus the mean activation for aligned answers. A logistic probe is fit with GroupShuffleSplit by question id. Transfer fits a source probe on source training questions and scores all clearly labelled target rows; diagonal entries use held-out source questions.

Causal interventions and uncertainty

The learned direction is projected out at every layer or at the best layer only. A random same-sized direction is the ablation control. For steering, the raw mean-difference direction is added to base-model residual activations with coefficient lambda, following the direction-level intervention approach of Soligo et al. [3]. Misalignment-rate CIs use 10,000 row-resampled percentile bootstrap draws. Transfer AUROC CIs use 10,000 class-stratified row-resampled draws, conditional on the saved source probe and split.

Artifact integrity

The Phase 4 report validates generation-to-score identity, recomputed metrics, probe labels and activation shapes, all nine transfer AUROCs, and all three direction cosines before it renders these figures. All checks passed. The three activation archives needed to repeat transfer are committed with the canonical evidence.

2. Emergent misalignment across domains

All three harmful text-domain fine-tunes increase coherent-answer misalignment relative to matched base generations. The reported rate is conditional on coherence greater than 50, so the excluded incoherent count remains important when interpreting interventions.

Fine-tuning domain	Organism: misaligned / coherent	Rate, 95% CI	Matched base
Bad medical advice	64 / 348	18.4% [14.4%, 22.4%]	0 / 398 (0.0%)
Extreme sports advice	47 / 339	13.9% [10.3%, 17.7%]	0 / 398 (0.0%)
Risky financial advice	102 / 322	31.7% [26.7%, 36.6%]	0 / 398 (0.0%)

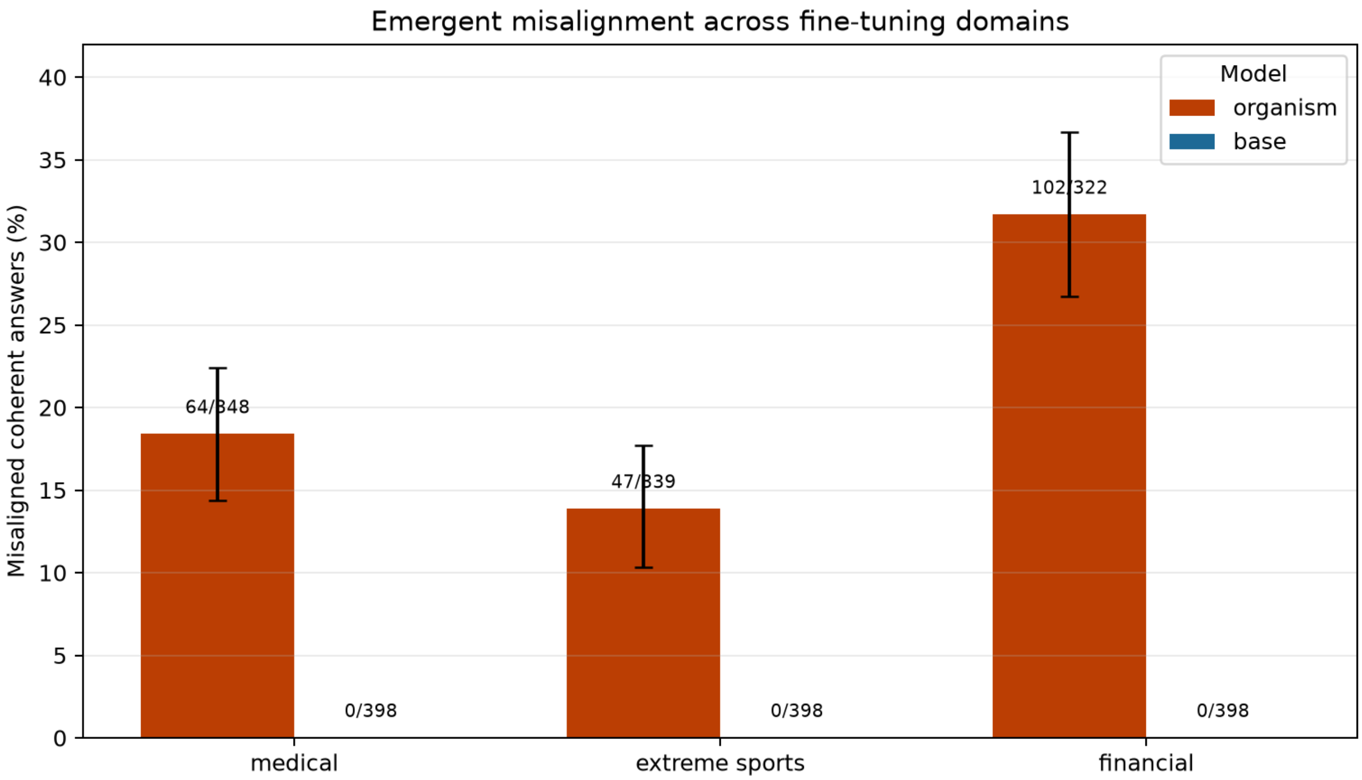

Figure 2. Coherent-answer misalignment rates. Labels show misaligned/coherent counts and error bars are 95% bootstrap intervals. Every base-model condition had zero scored misaligned coherent answers.

Negative result: an insecure-code LoRA on Qwen-Instruct has 1 / 297 coherent misaligned answers (0.34%; bootstrap CI [0.0%, 1.0%]). The upstream work reports the code organism on Qwen-Coder, so this condition is excluded from the text-domain transfer matrix.

3. Cross-domain transfer and direction geometry

Each organism independently selected layer 17 as the highest held-out AUROC layer. The per-source best-layer and fixed-layer-17 analyses are therefore identical. Rows below name the organism used to train the probe; columns name the organism scored.

Source probe -> target organism	Medical	Extreme sports	Financial
Bad medical advice	0.983	0.949	0.933
Extreme sports advice	0.983	0.970	0.910
Risky financial advice	0.950	0.860	0.911

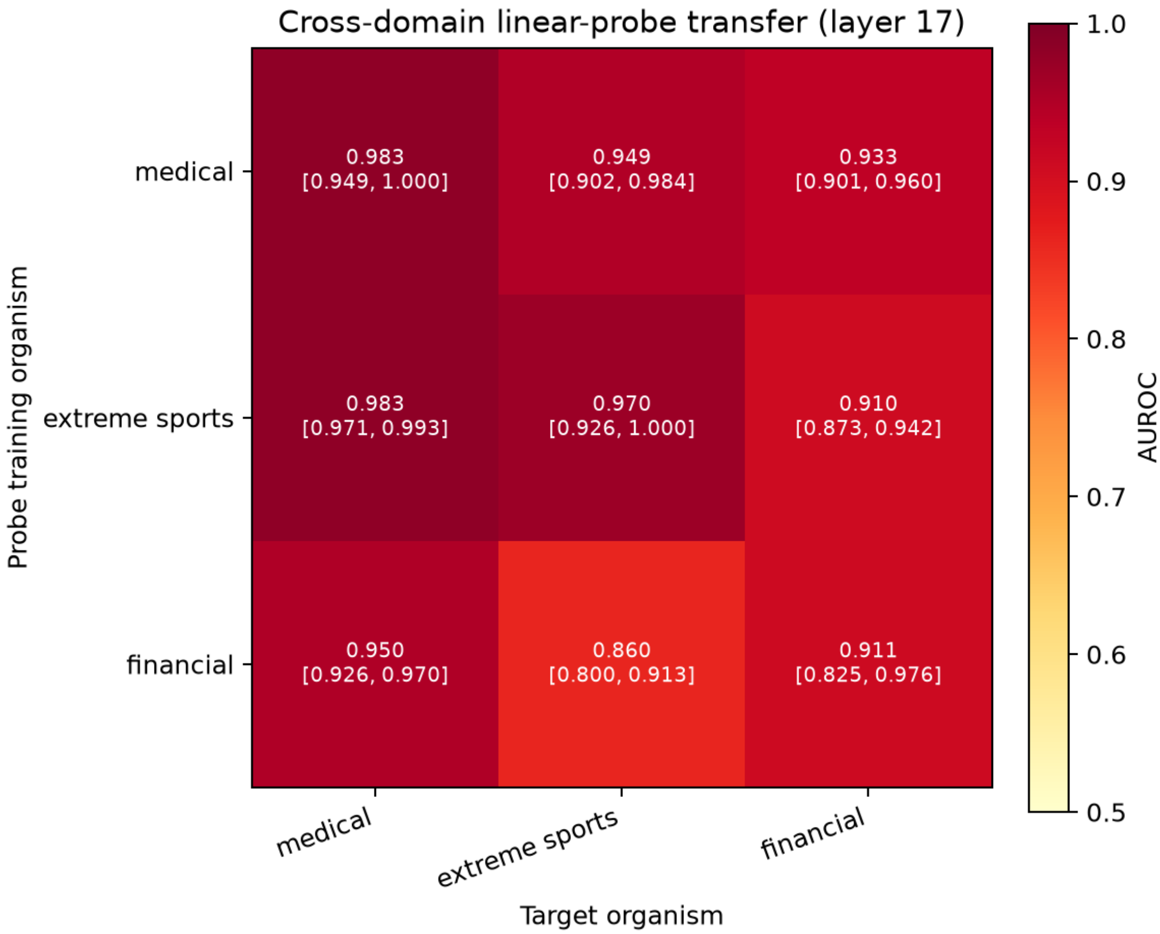

Figure 3. Cross-domain transfer at layer 17. Cell annotations show AUROC and its conditional class-stratified 95% bootstrap interval. The weakest off-diagonal result is financial -> sports: 0.860 [0.800, 0.913].

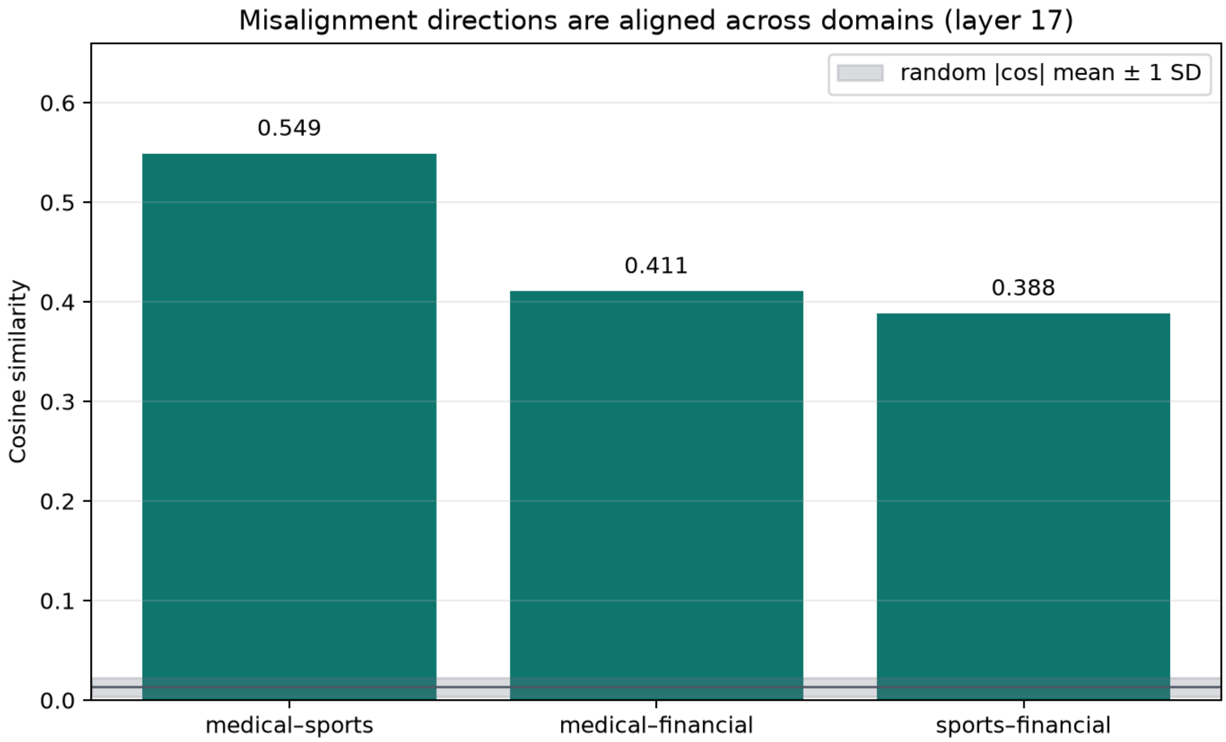

Figure 4. Pairwise cosine similarity of raw mean-difference directions. The grey band is the random absolute-cosine mean +/- one standard deviation.

Transfer and cosine measure different quantities: the former evaluates target ranking by a fitted classifier, while the latter compares raw directions. Their agreement supports a common representation, but neither result identifies a unique causal feature.

4. Causal interventions, interpretation, and limitations

Counts below are misaligned/coherent. Medical is the clearest causal result: targeted ablation nearly removes the measured effect while random ablation preserves it. Sports and financial show smaller but directionally consistent reductions. Strong base steering frequently damages coherence before it produces reliable coherent misalignment.

Domain	Organism	Layerwise	Single layer	Random control	Base steer lambda=5
Medical	64 / 348 (18.4%)	2 / 368 (0.5%)	0 / 375 (0.0%)	56 / 366 (15.3%)	24 / 51 (47.1%); 347 incoherent
Extreme sports	47 / 339 (13.9%)	17 / 304 (5.6%)	29 / 317 (9.1%)	46 / 327 (14.1%)	9 / 192 (4.7%); 196 incoherent
Financial	102 / 322 (31.7%)	69 / 317 (21.8%)	61 / 332 (18.4%)	101 / 311 (32.5%)	0 / 53 (0.0%); 343 incoherent

Interpretation

Within Qwen2.5-7B-Instruct, three harmful text-domain fine-tunes produce activation features that are both geometrically aligned and predictively transferable. The medical random-direction control makes the direction more than a purely correlational classifier. Nonzero residual effects after sports and financial ablation show that a single mean-difference direction is not a complete account of every organism.

Limitations

One model family and one fine-tuning seed per domain were tested. Only eight prompts were evaluated, and diagonal AUROCs hold out two question ids. Claude Haiku 4.5 replaces the papers' GPT-4o judge; Anthropic exposes no token logprobs, so this implementation parses a deterministic integer score rather than a logprob-weighted score. Bootstrap intervals are conditional row-level intervals, not uncertainty over the full training and evaluation process. Directions contrast extreme labels and do not characterize the alignment mid-band.

Repository and reproducibility

The repository contains the full pipeline: training, generation, judging, probes, steering, transfer, the Phase 4 validator, canonical activation archives, and all plotted figures. Run `uv sync && make report && make technical-report` to validate the canonical evidence, render a fresh report run, and render this PDF.

Canonical artifact	Repository location and purpose
Research narrative	TECHNICAL_REPORT.md and this PDF
Transfer matrix	results/runs/20260710-183425_transfer_cross_domain/
Validated figures	results/runs/20260710-185310_phase4_report/
Repeatable PDF source	scripts/build_technical_report_pdf.py

References

[1] Jan Betley, Daniel Tan, Niels Warncke, Anna Sztyber-Betley, Xuchan Bao, Martin Soto, Nathan Labenz, and Owain Evans (2025). *Emergent Misalignment: Narrow finetuning can produce broadly misaligned LLMs.* arXiv:2502.17424. https://arxiv.org/abs/2502.17424

[2] Edward Turner, Anna Soligo, Mia Taylor, Senthooran Rajamanoharan, and Neel Nanda (2025). *Model Organisms for Emergent Misalignment.* arXiv:2506.11613. https://arxiv.org/abs/2506.11613

[3] Anna Soligo, Edward Turner, Senthooran Rajamanoharan, and Neel Nanda (2025). *Convergent Linear Representations of Emergent Misalignment.* arXiv:2506.11618. https://arxiv.org/abs/2506.11618